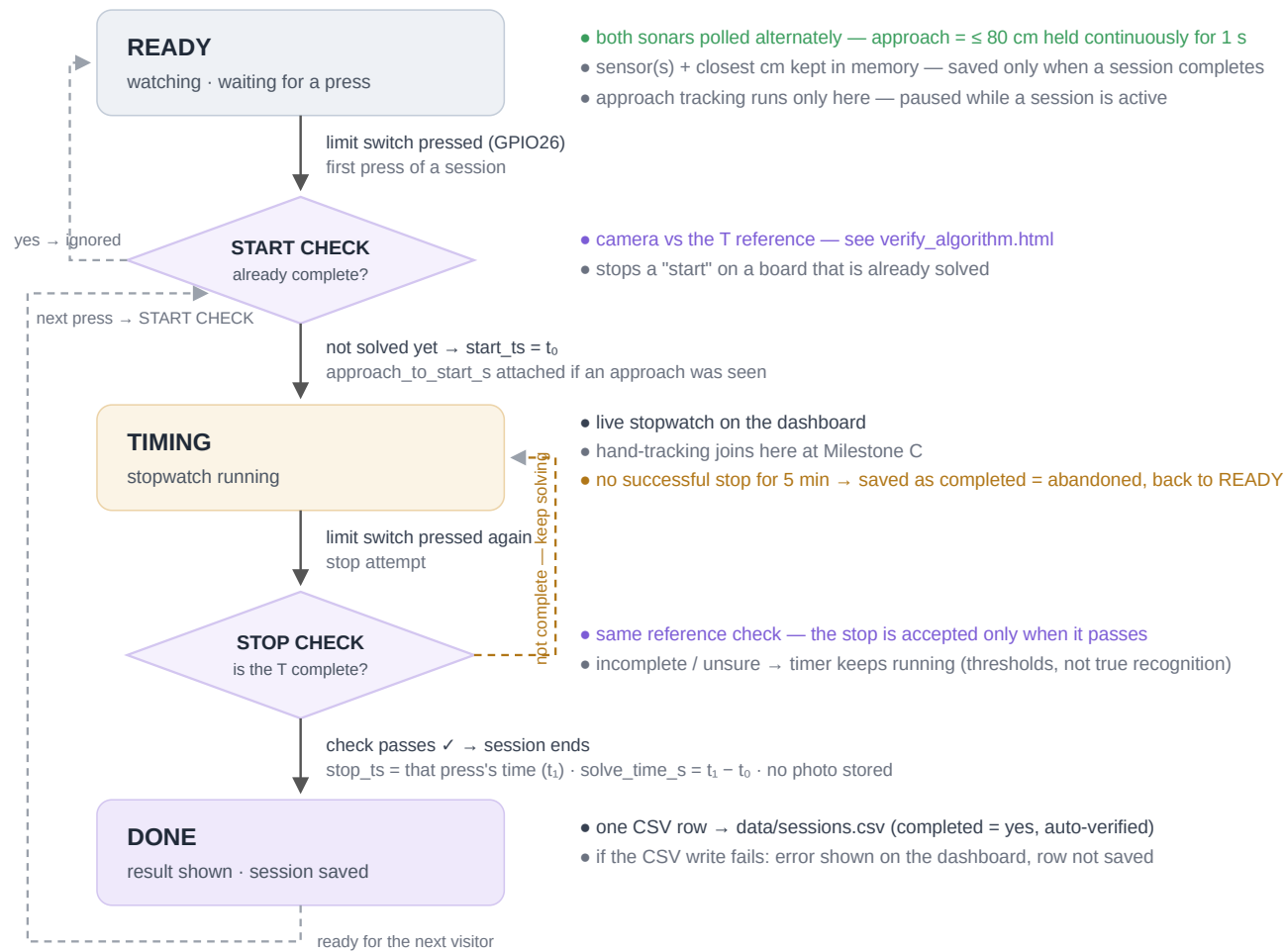


Puzzle Telemetry Rig — Phase 1

Param Foundation · AI & Robotics Lab · PFT/AIRL/CTX/2026-001 · rev 3, 20 Aug 2026 · pin map = docs/wiring.md rev 2 (probe-confirmed) · session flow = CLAUDE.md §4 rev 3

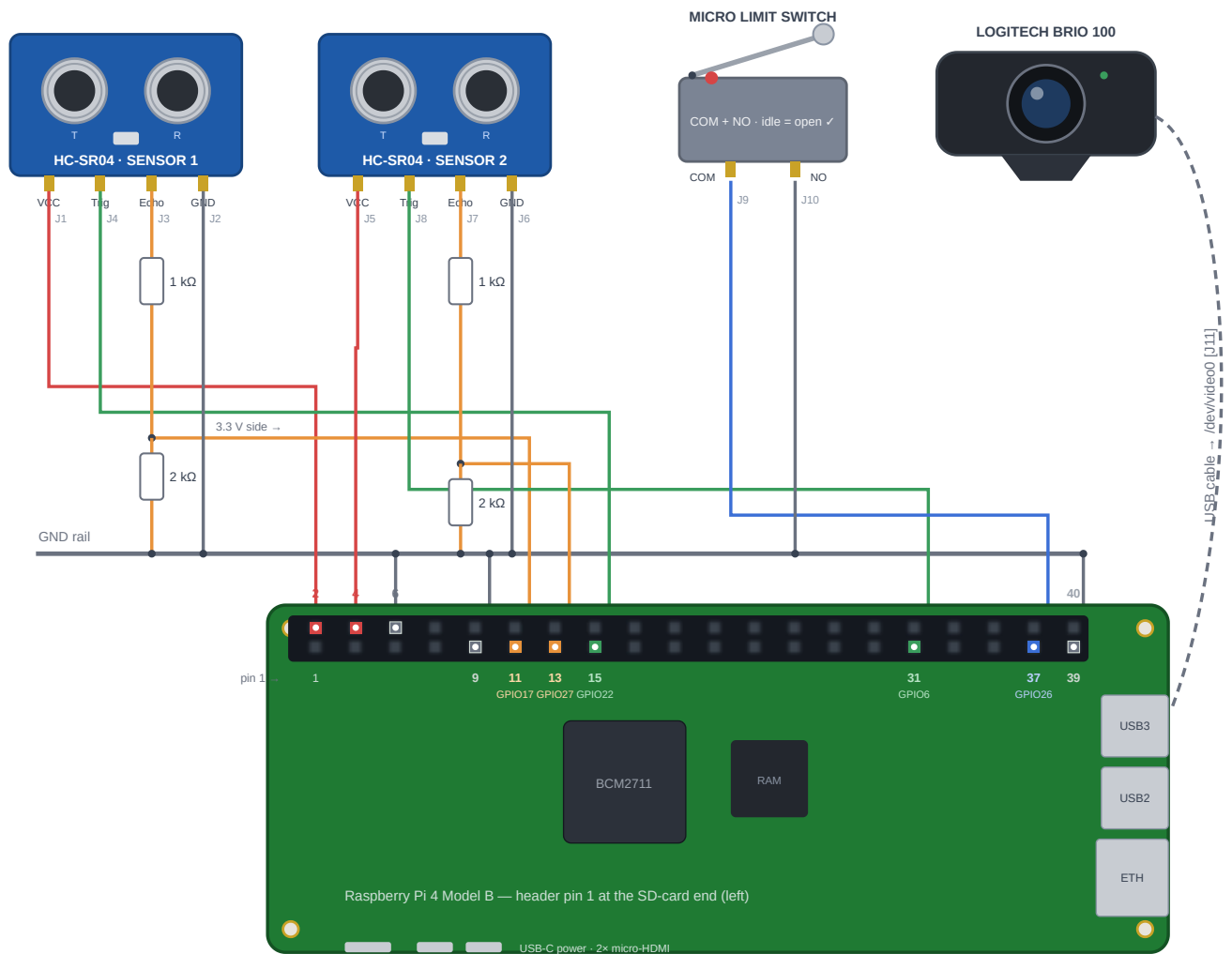
● 5 V power ● GND ● Trig (Pi → sensor) ● Echo (sensor → Pi, via 1 kΩ + 2 kΩ divider) ● limit switch --- USB ● camera completion check

1 · Session flow — press to start, press to stop · the camera check gates both



States from CLAUDE.md §4 (rev 3) · every threshold lives in src/config.yaml · purple = the completion check explained in [verify_algorithm.html](#)

2 · Bench wiring — every wire as it is physically connected (probe-confirmed)



- black dot = connected junction · a plain crossing is **not** a connection · J-refs match wiring.md §1 · both Echo lines pass through their 1 kΩ + 2 kΩ divider before touching the Pi (5 V → 3.3 V)

3 · Per-component wiring (probe-confirmed 20 Aug 2026)

HC-SR04 — Sensor 1		
Module	Pi pin	BCM
VCC	2 (5V)	—
Trig	15	GPIO22
Echo	11 · via divider	GPIO17
GND	6 (GND)	—

HC-SR04 — Sensor 2		
Module	Pi pin	BCM
VCC	4 (5V)	—
Trig	31	GPIO6
Echo	13 · via divider	GPIO27
GND	9 (GND)	—

Limit switch (start/stop)	
Terminal	Goes to
COM	pin 37 · GPIO26 (internal pull-up)
NO	GND rail

Idle = open, press = LOW — verified. Semantics: a press **starts** timing only if the T isn't already solved; a later press **stops** it only when the camera verifies completion.

E-paper 2.9" — deferred	
Conflict	Resolution when it arrives
GPIO17 = S1 Echo	move S1 Echo → GPIO23 (pin 16)

Then the display connects exactly as wiring.md §7 lists (SPI0, RST GPIO17, BUSY GPIO24). Driver `epd2in9_V2`, or legacy `epd2in9` for older IL3820 batches.